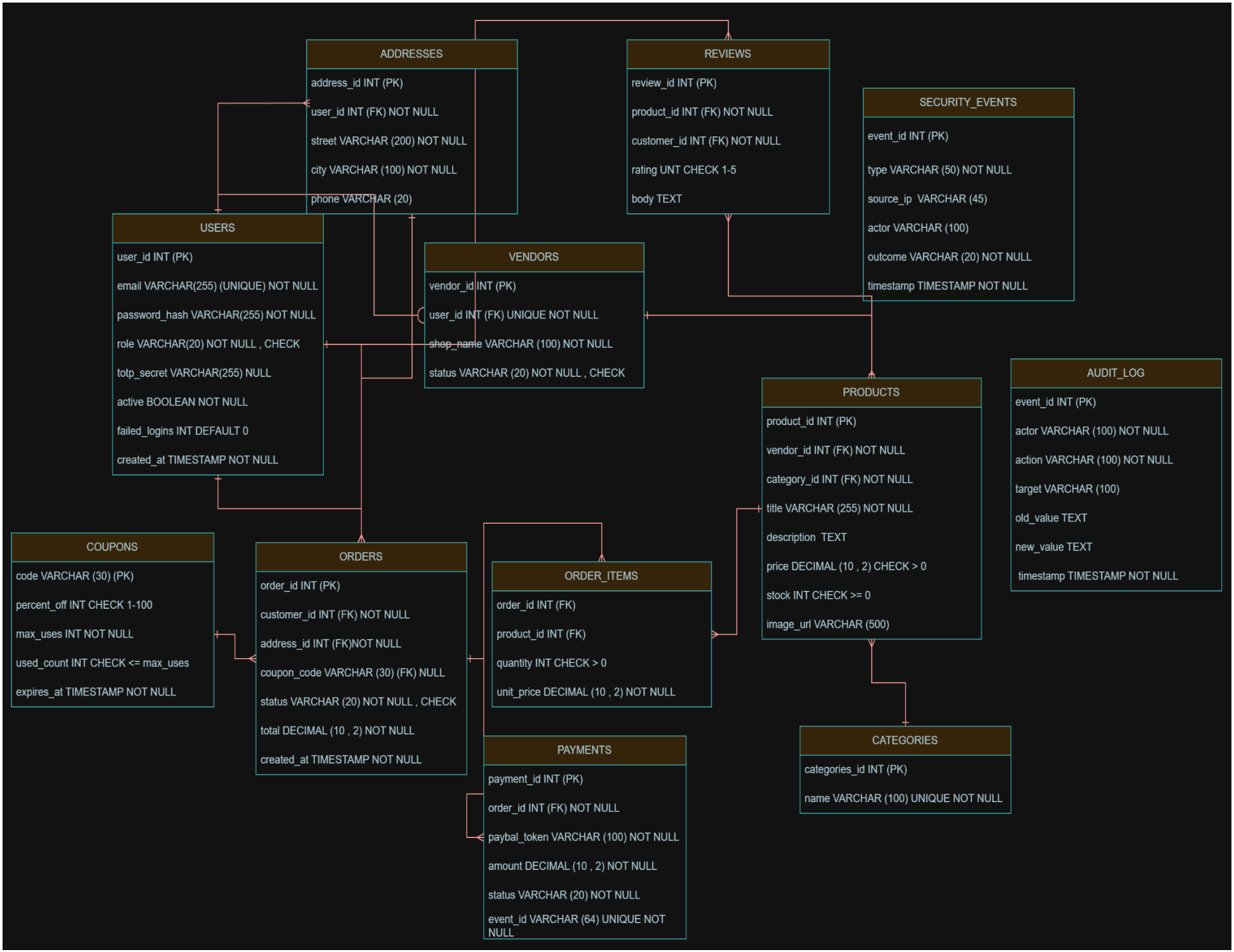

ADDRESSES
address_id INT (PK)
user_id INT (FK) NOT NULL
street VARCHAR (200) NOT NULL
city VARCHAR (100) NOT NULL
phone VARCHAR (20)
REVIEWS
review_id INT (PK)
product_id INT (FK) NOT NULL
customer_id INT (FK) NOT NULL
rating UNT CHECK 1-5
body TEXT
SECURITY_EVENTS
event_id INT (PK)
type VARCHAR (50) NOT NULL
source_ip VARCHAR (45)
actor VARCHAR (100)
outcome VARCHAR (20) NOT NULL
timestamp TIMESTAMP NOT NULL
USERS
user_id INT (PK)
email VARCHAR(255) (UNIQUE) NOT NULL
password_hash VARCHAR(255) NOT NULL
role VARCHAR(20) NOT NULL , CHECK
totp_secret VARCHAR(255) NULL
active BOOLEAN NOT NULL
failed_logins INT DEFAULT 0
created_at TIMESTAMP NOT NULL
VENDORS
vendor_id INT (PK)
user_id INT (FK) UNIQUE NOT NULL
shop_name VARCHAR (100) NOT NULL
status VARCHAR (20) NOT NULL , CHECK
PRODUCTS
product_id INT (PK)
vendor_id INT (FK) NOT NULL
category_id INT (FK) NOT NULL
title VARCHAR (255) NOT NULL
description TEXT
price DECIMAL (10 , 2) CHECK > 0
stock INT CHECK >= 0
image_url VARCHAR (500)
AUDIT_LOG
event_id INT (PK)
actor VARCHAR (100) NOT NULL
action VARCHAR (100) NOT NULL
target VARCHAR (100)
old_value TEXT
new_value TEXT
timestamp TIMESTAMP NOT NULL
COUPONS
code VARCHAR (30) (PK)
percent_off INT CHECK 1-100
max_uses INT NOT NULL
used_count INT CHECK <= max_uses
expires_at TIMESTAMP NOT NULL
ORDERS
order_id INT (PK)
customer_id INT (FK) NOT NULL
address_id INT (FK)NOT NULL
coupon_code VARCHAR (30) (FK) NULL
status VARCHAR (20) NOT NULL , CHECK
total DECIMAL (10 , 2) NOT NULL
created_at TIMESTAMP NOT NULL
ORDER_ITEMS
order_id INT (FK)
product_id INT (FK)
quantity INT CHECK > 0
unit_price DECIMAL (10 , 2) NOT NULL
PAYMENTS
payment_id INT (PK)
order_id INT (FK) NOT NULL
paybal_token VARCHAR (100) NOT NULL
amount DECIMAL (10 , 2) NOT NULL
status VARCHAR (20) NOT NULL
event_id VARCHAR (64) UNIQUE NOT NULL
CATEGORIES
categories_id INT (PK)
name VARCHAR (100) UNIQUE NOT NULL

Field	Level	Protection
title , price , image_url	public	No special protection
orders, order_items , coupons	Internal	Login and role checks
password_hash, totp_secret	Sensitive	Hashed or encrypted , never shown
street,city,phone	Sensitive	Visible to owner only
paypal_token	Sensitive	Never returned by the API